

What we're doing

Can independently replacing random subsets of a protein sequence with alanine produce complementary contacts-v1 rollouts that improve oracle best-of-100 accuracy on eval-val?

Why

Poly-alanine retains much of MarFold's learned contact-map prior, and bounded native/poly-alanine contrastive guidance improved oracle best@100 in #321. Sampling directly from independently partially alanine-mutated sequences is a much simpler way to perturb the conditional. The experiment asks whether the resulting hypotheses complement native rollouts at a fixed pool of 100.

Results so far

The intervention fails its development gate, so the untouched 81 proteins remain unread. Against the matched zero-mask worker, 5% masking lowers oracle best@100 by 0.0286 all-range and 0.0238 long-range; every larger rate is worse.

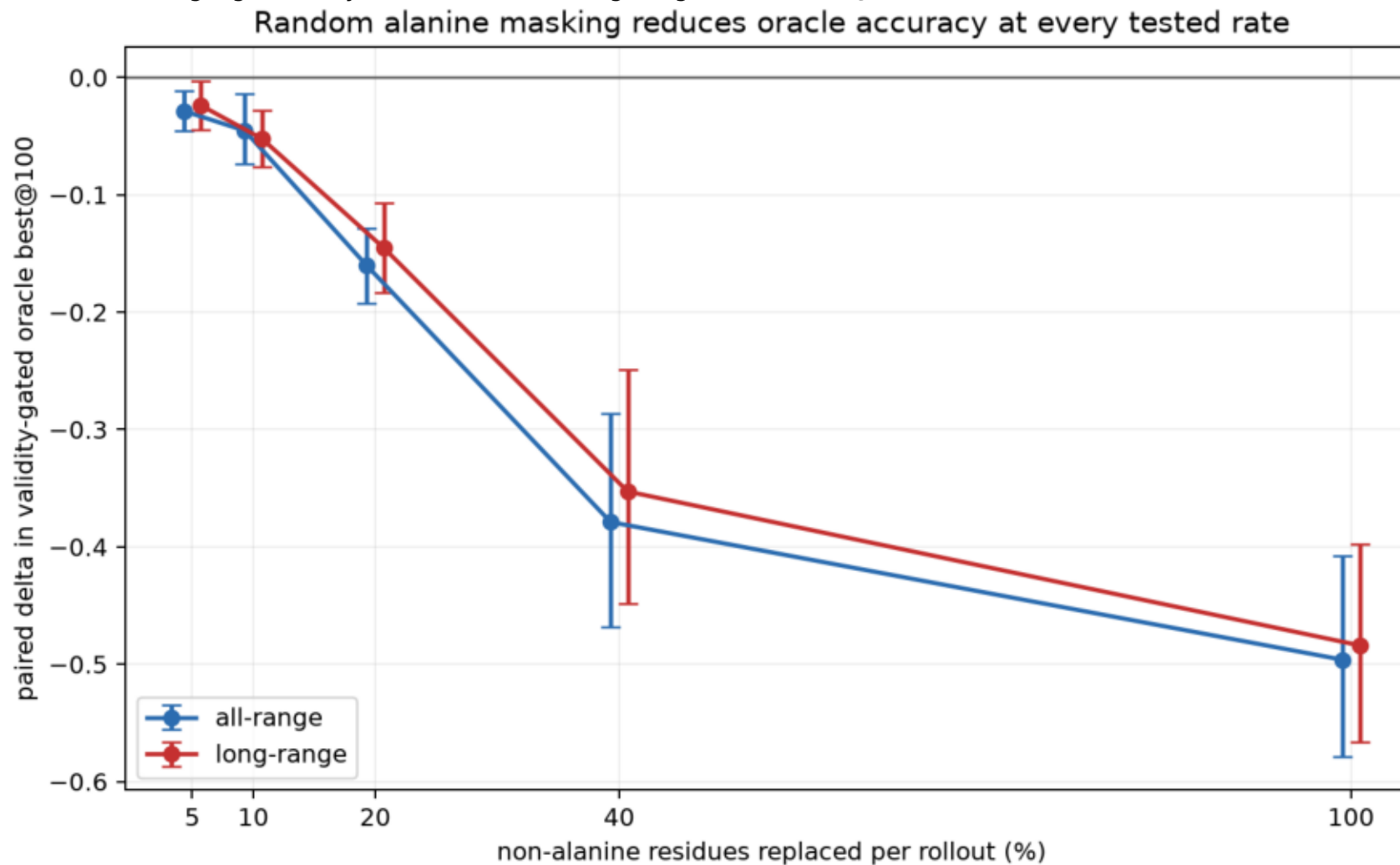
The best fixed-budget mixture (50 iid + 25 each at 5% and 10%) is +0.0079 all-range against the published iid100 pool but -0.0199 long-range. It therefore fails the co-primary long-range safety threshold. Masking lowers map overlap, but true-contact union recall and individual-map accuracy fall with it.

Conclusion

Do not advance random alanine masking to the 81-protein confirmation set. It creates novelty by erasing useful sequence evidence, not complementary accurate hypotheses. Full poly-alanine is degenerate: most rollouts emit only an end token and oracle accuracy is effectively zero.

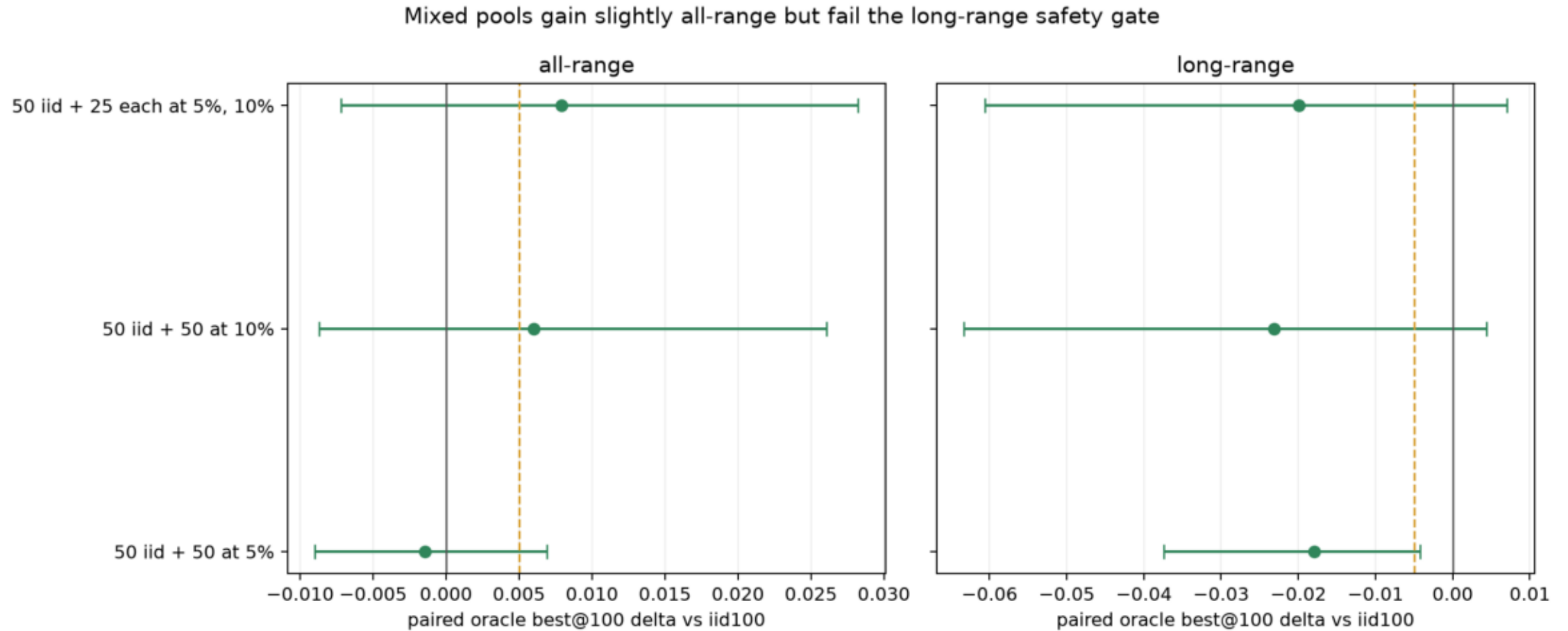
01_mutant_oracle_deltas.png

Paired protein-bootstrap mean deltas against the zero-mutation vLLM worker on the frozen 16-protein eval-val development set. Bars are 95% intervals. Even 5% masking significantly reduces all- and long-range oracle best@100.



02_mixed_pool_gate.png

Paired deltas against #321's published iid100 pool. Dashed lines mark the preregistered all-range improvement threshold (+0.005) and long-range safety threshold (-0.005). The two all-range-positive pools lose about two points long-range, so no policy advances.



03_diversity_tradeoff.png

All-range development means for 100 masked rollouts. Overlap falls through 40%, but true-contact union recall also falls. At 100% masking nearly every rollout ends immediately; its high empty-map Jaccard is a degenerate endpoint.

Masking creates different maps, but their true-contact coverage shrinks

